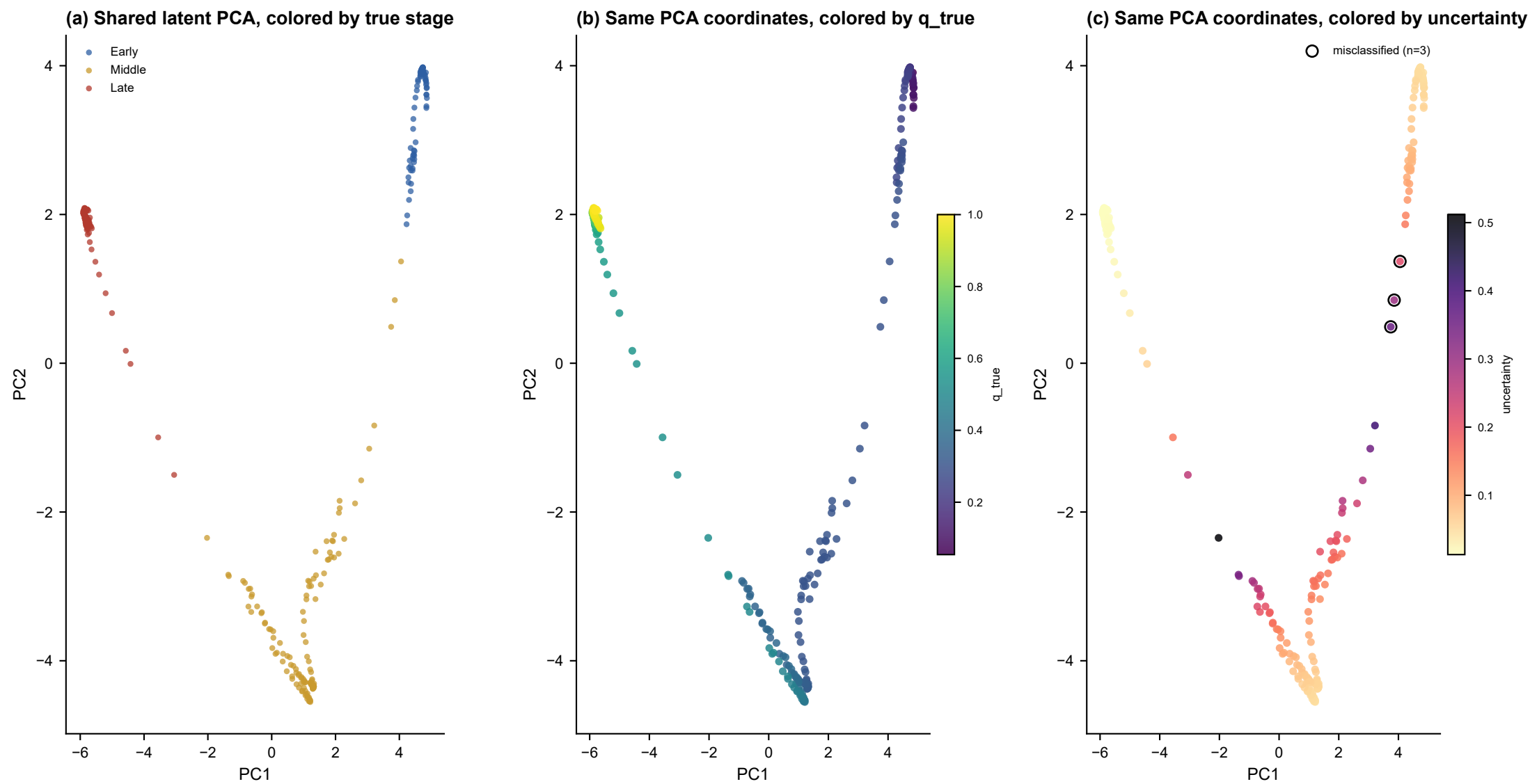
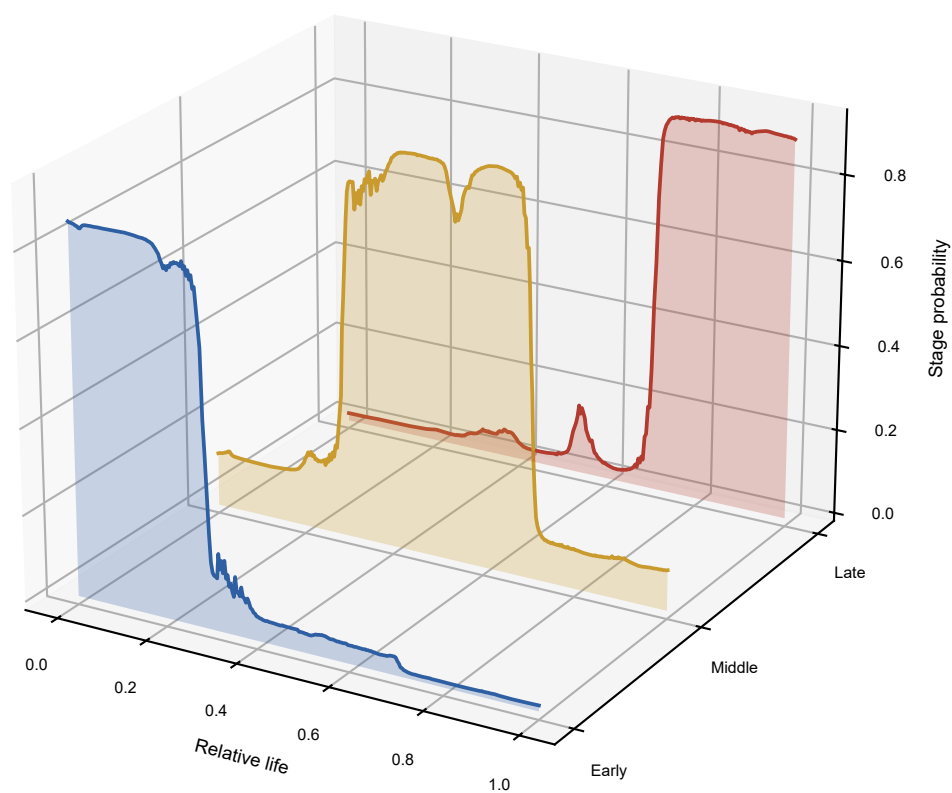


Shared 64-D latent representation geometry and joint probability–position evolution on PHM2010 D1/C6 (n=304): same PCA coordinates colored by stage/q/uncertainty, plus lifecycle stage-probability and confidence trajectories



(d) Stage-probability ridge lines over the C6 lifecycle



(e) Joint evolution: life $\times$ q_{pred} $\times$ confidence
(color = predicted stage; dashed = q_{pred} floor projection)

